When the Channel Changes

How the channels we use every day can strip emotion from speech, and what a compact AI model reveals about it

Jorge Mejia, PhD | Independent research | October 2026

The short version

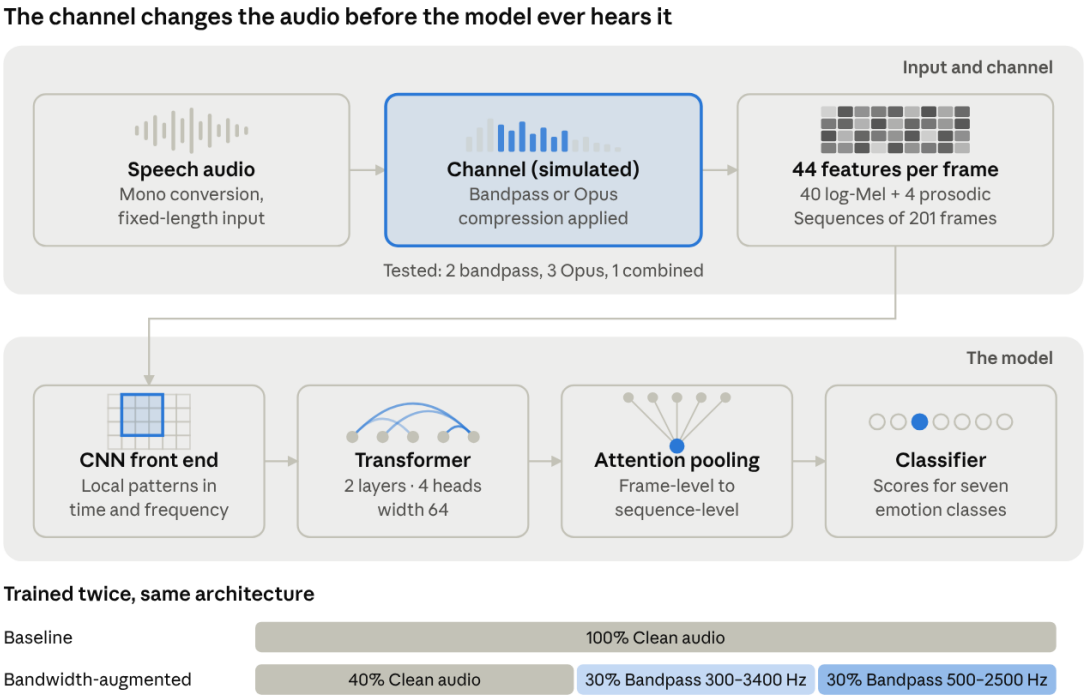
Every phone call, voice message and video meeting sends speech through a channel built to save bandwidth. Speech compression is typically optimised to preserve intelligibility, the words being said, but it can heavily alter the prosodic features, such as pitch movement, energy and voice quality, that carry emotion (Niebuhr and Siegert, 2023).

In listening studies of strongly compressed speech, emotions that should be easy to tell apart start to merge: high-arousal emotions such as joy were rated like boredom or sadness. That raises a question a machine can help test: how much emotional information survives the channels we use every day, and can it be protected? In this exploratory study, a compact emotion-recognition model lost more than half of its macro-F1 score (0.558 to 0.268) under a telephone-style frequency restriction. Training on restricted-bandwidth audio brought it back to 0.528, but the fix did not carry over to Opus compression.

Key takeaway: channels built to preserve words can degrade the cues that carry emotion, and the type of degradation matters, because bandpass limiting and compression behaved very differently. Knowing which cues are lost is the first step to designing channels and strategies that protect them.

Under the hood: how the model works

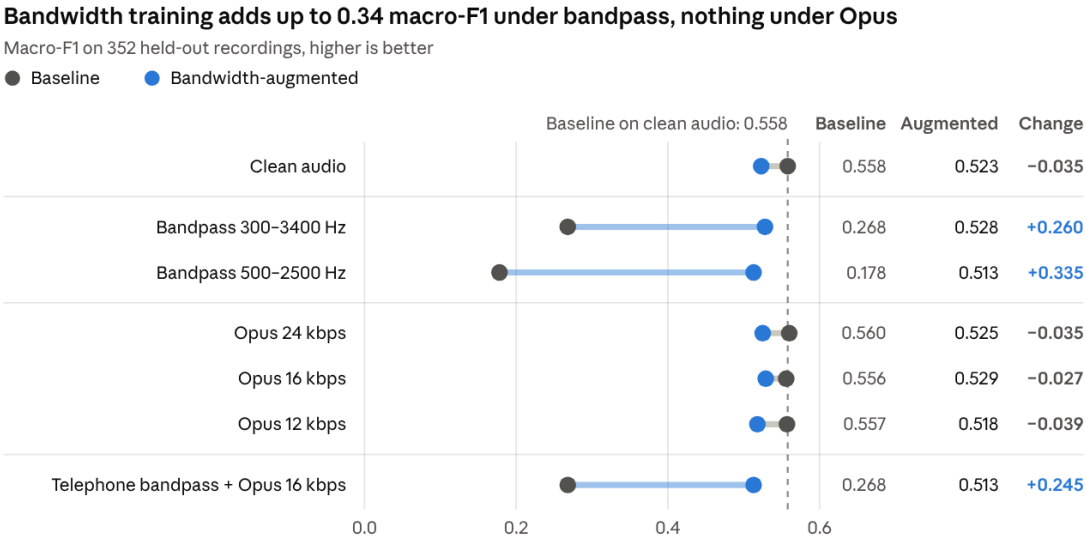
The classifier is a compact CNN–Transformer: a CNN picks out local time–frequency patterns, a Transformer relates them across time, and attention pooling condenses the clip into scores for seven emotions. It was trained on 1,372 acted-speech recordings and tested on 352 recordings from speakers it had never heard.



model schematic · 7 stages, 2 training mixes

Both models heard the same 352 held-out recordings through every channel, so differences come from training, not from the test.

What the results show



held-out test set, 352 recordings · one training run per model

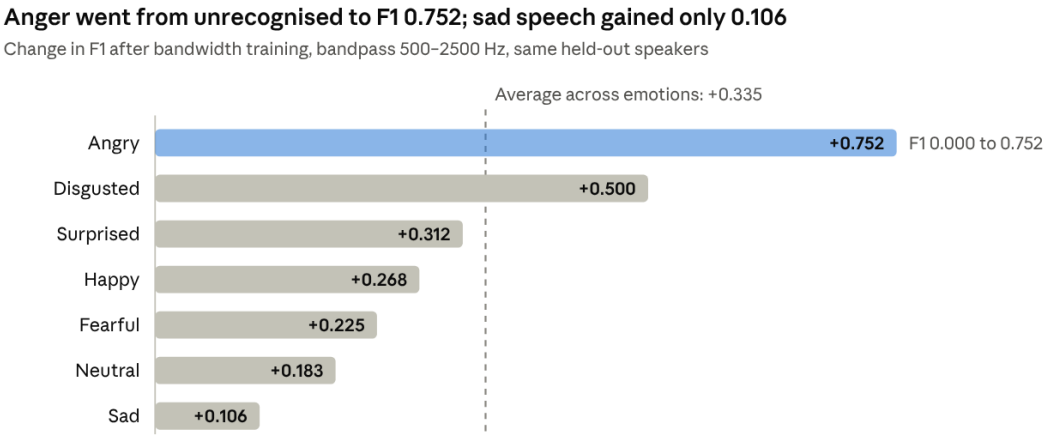
- **The baseline was fragile to frequency restriction.** Macro-F1 fell from 0.558 on clean audio to as low as 0.178 under a narrow bandpass.
- **Targeted training worked for the targeted problem.** The augmented model held at about 0.51 to 0.53 across the bandpass conditions, with bootstrap intervals above zero.
- **Compression was a different story.** Opus barely hurt either model, and augmentation did not help there.
- **There may be a small cost.** Clean-audio macro-F1 dipped from 0.558 to 0.523, though that interval includes zero.

Lessons for building audio AI

- **Test the channel, not just the clip.** A clean-audio score describes one situation. Build an evaluation suite that mirrors the transformations your system will actually meet.
- **Match augmentation to deployment.** Exposing a model to the degradation it will face recovered most of the lost performance, at far lower cost than a larger architecture.
- **Never assume robustness transfers.** Gains under bandpass filtering said nothing about codecs, so hold out unseen transformations and report each condition separately.
- **Keep the claim in proportion.** A compact model, a paired test set and bootstrap intervals support a specific claim, not general robustness and not human perception.

What the model tells us

Beyond the numbers, the model points to something fundamental: transmission channels can degrade the acoustic cues that carry emotion in speech, and not all emotions are affected equally. Traditional telephone quality stops at 3.4 kHz, and in earlier codec work the narrower codecs lost more of the acoustic separation between emotions, consistent with bandwidth being one cause (Niebuhr and Siegert, 2023). Under the narrow 500–2500 Hz bandpass here, the baseline failed to recognise a single angry recording, yet recovered to an F1 of 0.752 after bandwidth training, while sad speech improved by only 0.106. This gives an initial foundation for designing more robust transmission channels, and strategies that reduce the loss.



bandpass 500–2500 Hz · 352 held-out recordings · one training run per model

Sad speech also shows why recall alone is not enough: the baseline recalled 0.922 of sad recordings but raised many false alarms, so its F1 was only 0.340, while the augmented model's recall fell to 0.547 and its F1 rose to 0.446. These are model-specific findings, so human validation is needed before concluding that listeners would experience the same emotion-specific effects.

Limits of the model

The model is a specification, not a listener. It is unlikely that people follow a fundamentally different approach to perceiving emotion; the more likely difference is how the cues are weighted. The model also hears isolated clips, whereas people draw on context such as the conversation so far, the speaker and the situation, and its absence can change what is heard. With a single training run, one acted-emotion dataset and simulated channels, the results are evidence about this model under these conditions, not a measure of human perception.

Where next

A per-emotion analysis of which emotions lose the most, multiple training seeds, speaker-level resampling, an independent dataset and, with ethical oversight, a listening study with human participants.

Conclusion

Targeted training augmentation can improve robustness to the bandwidth restrictions evaluated here, but the benefit depends on the type of channel degradation. For anyone building or buying audio AI, the practical question is not "how accurate is it?" but "accurate under which conditions, and how do you know?"

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